

CSC 411 ML Lecture 11

The Transition to Industry ML

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12th May 2026
SST Lab 2

Brief Overview

- ❑ MLOps, CD4ML, Docker, TFX, Big Data, and Post-Deployment Drift.
- ❑ Bias Mitigation, Human-in-the-Loop (HITL), and High-Stakes Accountability.

● Recap of The ML Workflow

Problem Formulation

Gather and organize data

Preprocessing, cleaning, and visualizing

Choose a model and optimize

Training: Fit a new model on the training set.

Linear Models & Regularization

Non-Parametric Models & Ensembles

Support Vector Machines

Unsupervised Learning

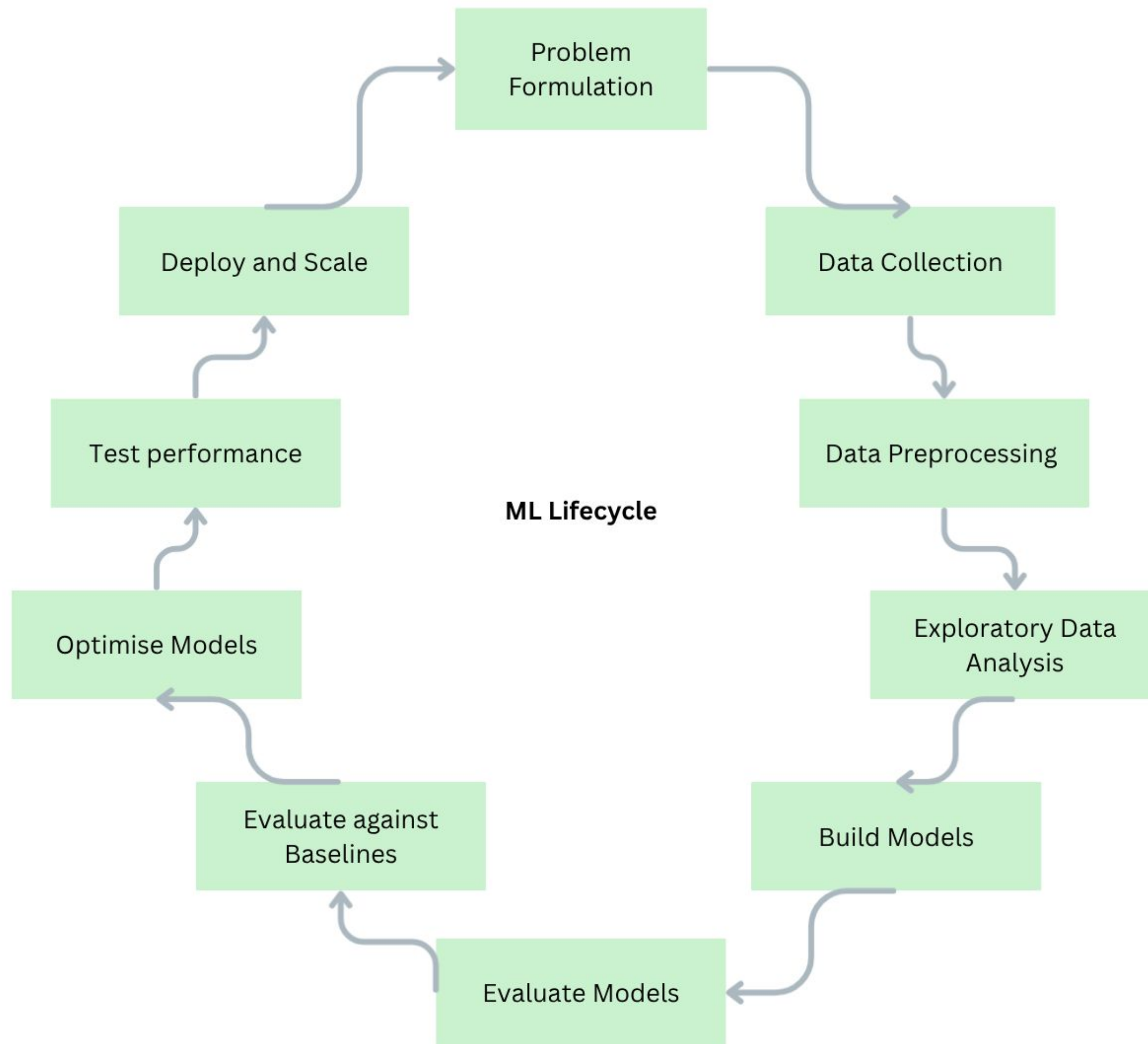
Reinforcement Learning

Model Selection, Evaluation, and Learning Theory

Model Selection and Tuning

MLOPs and Deployment

Monitoring and Improvement



● Hidden Technical Debt in ML

Training a model is only a **tiny** fraction of the work.

There is a vast array of surrounding **infrastructure** and **processes** to support model evolution.

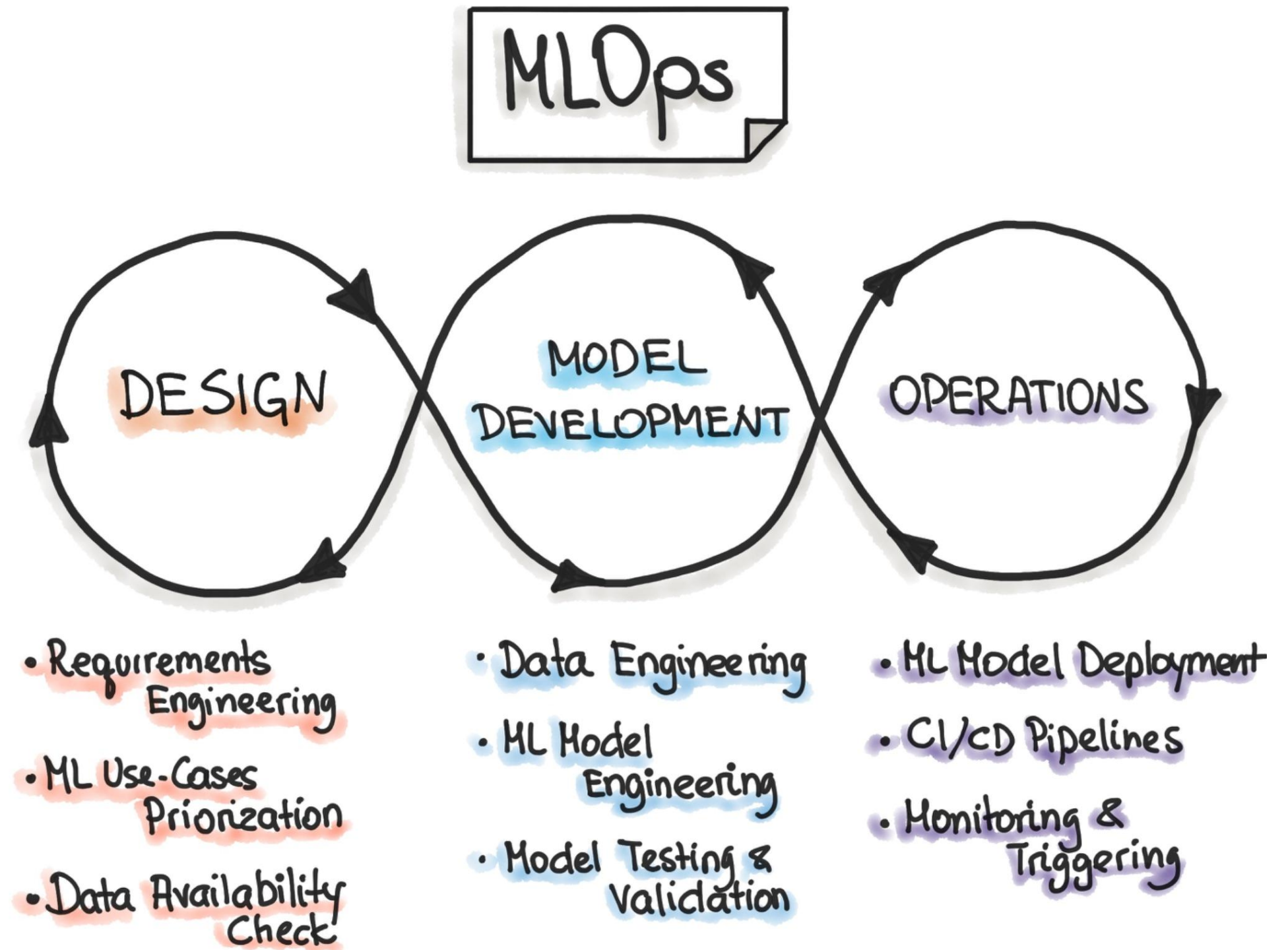
● MLOps and Continuous Delivery for ML (CD4ML)

What is MLOps?

A set of practices at the intersection of Machine Learning, DevOps, and Data Engineering.

Goal:

- Deploy and maintain ML models reliably and efficiently.



Source: ML-OPs.org

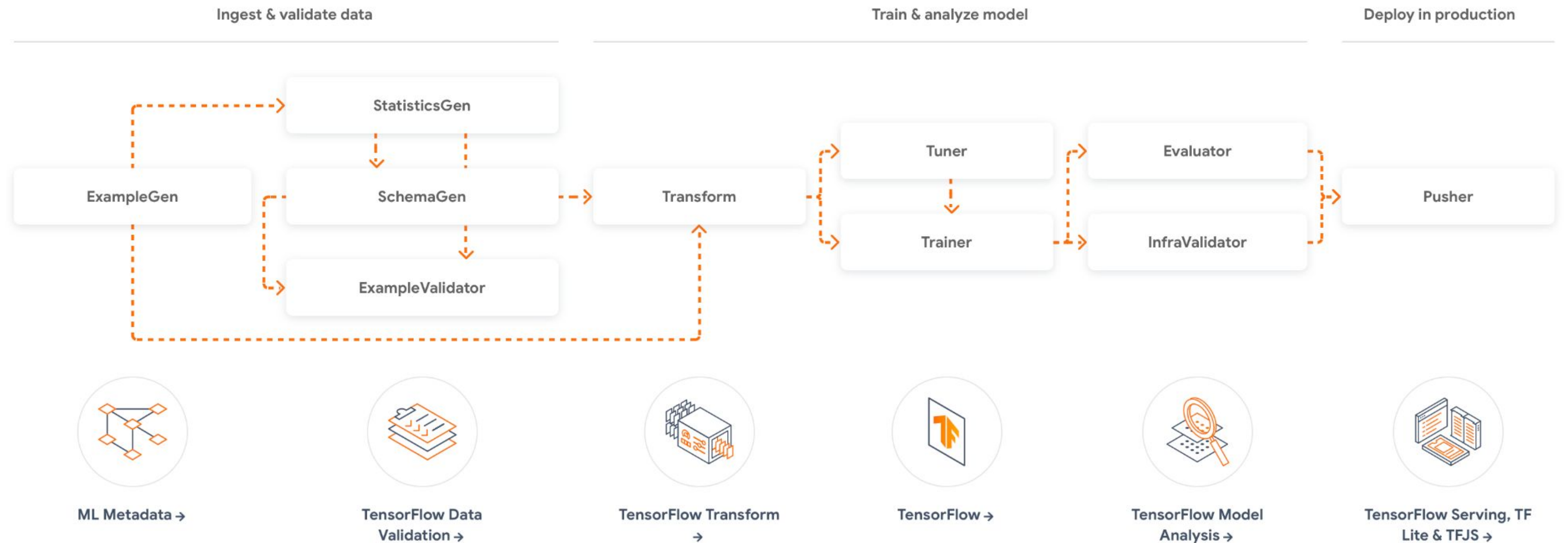
● Continuous Delivery for ML (CD4ML)

- Extends traditional Continuous Delivery to ML systems.
- Manages the 3 axes of change: Code, Model, and Data.

Automating ML Pipelines:

- Manual pipelines are prone to inefficiencies and scaling difficulties.
- Frameworks like Kubeflow and TensorFlow Extended (TFX) automate workflows in cloud-native environments.

● The Anatomy of a TFX Pipeline



Source: TFX website

● The Anatomy of a TFX Pipeline

ExampleGen

Ingests data.

Transform

Performs feature engineering.

Trainer & Tuner

Trains models and optimizes hyperparameters.

Evaluator & Pusher

Assesses performance and deploys validated models.

● The "It Works on My Machine" Problem

A model trains perfectly on your laptop but crashes on the production server.

Cause:

- ❑ Different OS
- ❑ Missing dependencies
- ❑ Conflicting library versions.
- ❑ Data Management, etc

● Containerizing Intelligence with Docker

Docker: Packages software into standardized units called containers.

Docker provides CI/CD Guarantee

Containers: include everything needed to run: code, runtime, system tools, and libraries.

Containers vs. Virtual Machines (VMs)

- ❑ VMs: Heavy, includes a full guest Operating System.
- ❑ Containers: Lightweight, shares the host OS kernel, starts in milliseconds.

● Infrastructure and Cloud Computing

Infrastructure as Code (IaC)

- ❑ Managing and provisioning computing data centers through machine-readable definition files.
- ❑ Replaces physical hardware configuration and interactive menus.

Managing ML Infrastructure with Terraform

The industry standard for IaC is Terraform

Terraform: An open-source IaC software tool.

- ❑ Defines cloud resources safely and predictably; integrates with Jenkins and Ansible

Industry ML requires **massive** storage and GPU compute power.

● The MLOPs Need

Unified MLOps & Data Architectures

- Siloed ML systems lead to redundant work, slow deployments, and operational overhead.
- Unified architectures enable shared Feature Stores and consistent deployment.

Scalability and Elasticity

- Scalability: Handling growing amounts of work by adding resources.
- Elasticity: Automatically scaling resources up or down based on real-time demand.

● Big Data Frameworks

The Big Data Challenge

- ML models (especially deep learning) require massive datasets.
- The 3 Vs: **Volume**, **Velocity**, **Variety**.
- Datasets often exceed the memory (RAM) of a single machine.

Distributed Computing

- Instead of buying one massive supercomputer, network thousands of cheap commodity computers together.
- Distribute the data and the processing across the cluster.

Distributed computing is useful for training, inference serving, and reliability

● Big Data Frameworks



Apache Hadoop

- An open-source framework for distributed storage and processing.

It has two main parts:

- HDFS: Hadoop Distributed File System.
- MapReduce: Programming model for parallel processing.

HDFS shatters massive datasets into blocks and scatters them across the cluster for storage

MapReduce is a programming model that sends the ML algorithm to the data, rather than moving the data to the algorithm.

● Big Data Frameworks



Apache Spark

- A unified analytics engine for large-scale data processing.
- Keeps data in-memory (RAM) for blazing-fast iterative computation.

While **Hadoop** writes intermediate results to physical hard drives, **Apache Spark** keeps data in-memory (in RAM).

Because it avoids slow hard-drive read/writes, **Spark** is uniquely suited for machine learning algorithms

● Big Data Frameworks



Spark MLlib

- Spark is up to 100x faster than Hadoop MapReduce for iterative ML algorithms.
- Provides distributed implementations of classification, regression, and clustering algorithms.

Spark includes **MLlib**, a library containing distributed versions of the algorithms we learned earlier in the semester, allowing you to train models on petabytes of data.

● Post-Deployment Drift & Human-in-the-Loop

Deployment is Just the Beginning

- A deployed ML model is a living system.
- Real-world data is dynamic; most failures occur after deployment.

Data Drift

- Changes in the statistical properties of data over time.
- The model's underlying mapping is still correct, but the input distributions look different.

Concept Drift

- Changes in the relationship between input features and the target variable.
- The very definition of what you are trying to predict has shifted.

● Post-Deployment Drift & Human-in-the-Loop

Monitoring and Mitigation

- **Proactive Monitoring:** Early warnings for pipeline drift.
- **Metrics:** Track both quantitative and qualitative metrics
- **Incremental Updates:** Adopt online learning to update models without full retraining.
- Implement **A/B tests** to observe feature changes and improvements

Human-in-the-Loop (HITL) Systems

Humans intervene in ambiguous, biased, or dynamic environments.

- **Active Learning:** The model identifies data points it is uncertain about and queries humans for labels.
- **Crowdsourcing:** Scaling human input cost-effectively for micro-tasks and moderation.

● Part B: Ethics, Fairness, and Bias Mitigation

With great predictive power comes massive societal responsibility.

Artificial Intelligence Risk Management (AI RMF)

- A framework (by the US gov) for identifying, managing, and mitigating AI risks.
- Connects technical system design to organizational values and principles.

ML as a Mirror of Society

- If the world has a problem, the data will reflect it, and the model will learn it.
- The Danger of the Feedback Loop:

**Biased predictions generate new
biased data!**

● Part B: Ethics, Fairness, and Bias Mitigation

Where does Bias come from?

- **Representation Bias:** Training data does not accurately represent the population.
- **Model Bias:** Algorithms learning toxic associations (e.g., word embeddings linking gender to professions).
- **Proxy Variables:** Using "zip code" as a proxy for race or socioeconomic status.
- **Annotation bias:** culture, ideology, and lack of shared standards

Bias Mitigation: Pre-processing

- Asking the hard questions to stakeholders
- Fixing the data before training.
- Relabelling (Massaging): Switching labels of instances near the decision boundary.
- Sampling/Reweighting: Assigning weights to instances based on label and protected attribute.

● Part B: Ethics, Fairness, and Bias Mitigation

Bias Mitigation: In-processing

- Reassessing model choice.
- Modifying the algorithm to capture missing link
- Adding mathematical fairness constraints or regularization penalties to the cost function.

Bias Mitigation: Post-processing

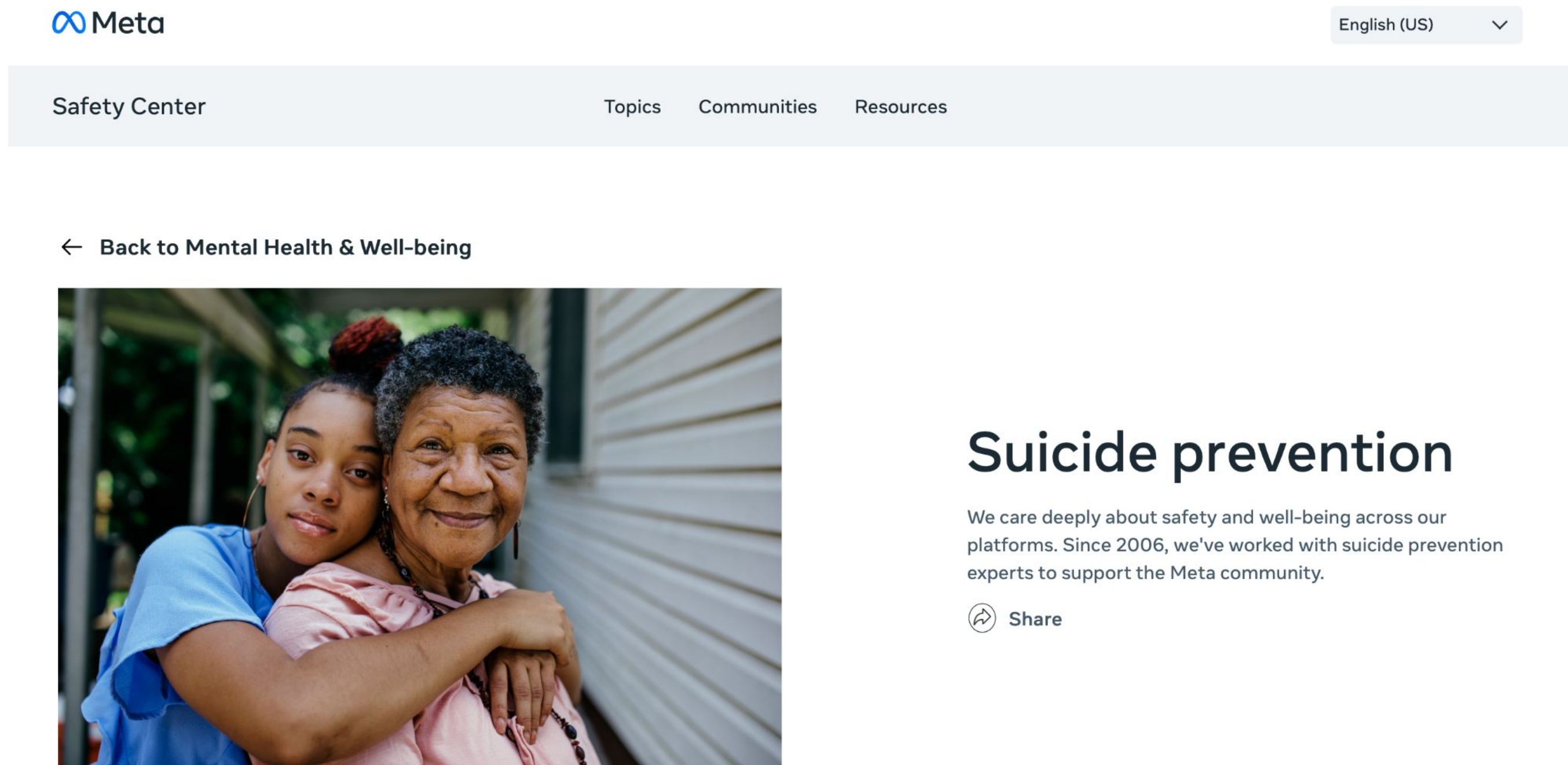
- Robust testing and evaluation
- Mathematically adjusting the predictions (using evolutionary algorithms)
- Adjusting the decision threshold for different groups to achieve parity.

The Fairness vs. Accuracy Trade-Off

Forcing a model to adhere to strict fairness constraints often decreases its overall statistical accuracy.

● ML in High-Stakes Settings

Case Study 1: Facebook's Suicide Prevention



● ML in High-Stakes Settings

Ethical Dilemmas in the Facebook Case

- **Pros:** Rapid intervention saves lives at scale.
- **Cons:** Massive privacy invasion, potential for false positives leading to unwarranted police interventions.

● ML in High-Stakes Settings

Case Study 2: The "Hungry Judge" Effect. Algorithms can unintentionally learn and amplify these hidden human biases.



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Hungry judge effect

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From Wikipedia, the free encyclopedia

The **hungry judge effect** is the observation that judges' verdicts are more lenient after a meal break. Since the original study, the term has morphed to encompass a stream of research concerned with implications of hunger on economic and social behavior. However, it has been suggested that this may be an artifact of case scheduling.^[1]

Original study [\[edit \]](#)

A study of the decisions of Israeli [parole boards](#) was made in 2011.^[2] It found that the granting of parole was 65% at the start of a session but would drop to nearly zero before a meal break.^[2] The authors suggested that mental depletion as a result of fatigue caused decisions to increasingly favour the status quo, while rest and replenishment then restored a willingness to make bold decisions. The paper, which was published in the *Proceedings of the National Academy of Sciences*, has been [cited many times](#) – 1,380 times by 2021.^[3]

Thank you

Questions?